

MD AJMEER KHAJA

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Education

Osmania University, Bachelors in Computer Science – Hyderabad, Telangana July 2020 – July 2023
State Board Of Technical Education & Training, Diploma in Electrical and Electorics June 2017 – Apr 2020
– Mahabubnagar, Telangana

Experience

Senior AI Engineer, INNOVAPATH IT Solutions PVT Ltd – Hyderabad,Telangana June 2025 – present

- Designed and deployed an AI-driven analytics platform enabling natural language queries to SQL and graph databases, improving data retrieval efficiency and increasing adoption among non-technical users.
- Built a dashboard generation engine using LLMs to convert user queries into structured JSON visualization configs, supporting real-time rendering and significantly reducing manual dashboard creation effort.
- Developed a contract data processing pipeline on AWS (S3, Lambda, Textract/Comprehend or similar) to extract entities and relationships from unstructured documents, improving downstream healthcare analytics accuracy and reducing manual review effort.
- Architected and deployed FastAPI-based microservices on AWS EKS, implementing RBAC, logging, and monitoring with Prometheus/Grafana, while optimizing performance through caching and query tuning.

AI Engineer, INNOVAPATH IT Solutions PVT Ltd – Hyderabad,Telangana Feb 2024 – May 2025

- Built a Retrieval-Augmented Generation (RAG) assistant to automate candidate shortlisting by matching resumes to job descriptions based on skills, experience, and location, reducing manual screening effort by ~40%.
- Designed and deployed a document ingestion pipeline using AWS (S3, Lambda, DLQs) and Docker to process and index 10K+ resumes and job descriptions with improved reliability and fault tolerance.
- Implemented a hybrid search system combining vector similarity (Sentence Transformers, Milvus) with keyword-based retrieval, increasing candidate-job match accuracy by 25% compared to keyword-only search.
- Developed LLM workflows using LangChain and AWS Bedrock for query refinement and response generation, enabling recruiters to retrieve ranked candidates with source-backed explanations, reducing turnaround time per search by 30%.

AI ML Engineer Intern, INNOVAPATH IT Solutions PVT Ltd – Hyderabad,Telangana Sept 2023 – Jan 2024

- Assisted in developing REST APIs using FastAPI for ML model integration and backend services.
- Contributed to CI/CD workflows using GitHub Actions for testing and deployment automation.
- Helped implement Redis caching for frequently accessed endpoints to improve response times.
- Worked with Docker containers and assisted in deployments on Amazon Web Services Amazon EKS environments.

Projects

Confidential Healthcare Agentic AI system

Agentic AI systems, MCP servers, and LookML-driven analytics; built end-to-end platforms for natural language querying, automated insights, and dashboard generation.

- Designed and deployed an agentic AI system enabling natural language → SQL/graph queries, improving data access for non-technical users.
- Built an LLM-driven dashboard generation engine capable of dynamically constructing structured outputs (JSON specs) for visualization layers, enabling real-time dashboard rendering
- Built a contract intelligence pipeline ingesting documents from Amazon S3, extracting structured entities (contracts, amendments, payers), and linking relationships for downstream analytics.
- Modeled complex healthcare data relationships in Neo4j by building an MCP server with 20+ modular tools, enabling efficient graph-based querying.
- Built a context management + memory layer (session memory, retrieval cache, embeddings) to support multi-turn conversations and personalization

- Exposed capabilities via FastAPI-based microservices, deployed on AWS (EKS), with support for scalable inference and async processing
- Implemented RBAC-based authentication, observability (logging, tracing), and fault-tolerant workflows to ensure production reliability.
- Optimized system performance across latency, token usage, and retrieval accuracy.

TalentScreen Agentic RAG system (Phase 2)

Extended an existing Retrieval-Augmented Generation (RAG) system into a multi-agent architecture using LangGraph to support automated resume evaluation and candidate communication workflows.

- Designed and implemented multiple task-specific agents, including resume comparison, review insights generation, recommendation generation, and automated email response handling.
- Built an orchestration layer using a supervisor agent pattern to coordinate agent interactions, manage task flow, and handle fallback scenarios.
- Implemented short-term and long-term memory management for agents, enabling context retention across sessions and improving response consistency.
- Addressed prompt reliability and output consistency issues through iterative prompt tuning and structured response handling.

TalentScreen RAG system

RAG based AI assistant for Talent Management for recruiter.

- Built a Retrieval-Augmented Generation (RAG)-based AI assistant to support talent management and recruiter workflows.
- Developed a scalable data ingestion pipeline using AWS S3, AWS Lambda, Docker, and Dead Letter Queues (DLQs) for robust document processing.
- Leveraged LangChain and AWS Bedrock to orchestrate LLM workflows and enable context-aware query handling.
- Implemented document preprocessing and metadata extraction, generating embeddings using Sentence Transformers and storing them in a Milvus vector database.
- Designed a hybrid retrieval system combining semantic search (vector embeddings) and keyword-based methods to improve relevance.
- Integrated citation generation to provide traceable and explainable responses.
- Applied query optimization techniques to better interpret complex and ambiguous job descriptions.

White-box learning Backend

Worked on backend systems for a US-based Learning Management System (LMS), handling API development, access control, and scalable deployment for a multi-role platform (Admin, Staff, Learners).

- Developed REST APIs with FastAPI using async handling and optimized queries.
- Added Redis caching for high-read endpoints, improving response times and reducing DB load.
- Implemented RBAC with JWT authentication and middleware-based access control.
- Built CI/CD pipelines with GitHub Actions for automated testing and deployments.
- Containerized services with Docker and deployed on AWS EKS with autoscaling and rolling updates.

Skills

Programming Languages: Python, JavaScript

Frameworks: Pytorch, MLflow, Reactjs, Nodejs, Redux

Libraries: Langchain, Llamaindex, NLTK, HuggingFace, Transformers, Scikit-learn, NumPy, Pandas

LLMs: LLaMA, Anthropic Claude, BERT, Mistral, OpenAI

Database: SQL, PostgreSQL, MongoDB

AWS: EKS, Bedrock, RDS, Lambda, CloudWatch, S3 bucket, EC2, API Gateway